ATIVIDADE DE BANCO DE DADOS

ETEC Irmã Agostina

Grupo: Gabriel Dantas, Nicole Faustino, Thifany Godoi e Guilherme Borges

Este documento apresenta a especificação física do banco de dados relacional desenvolvido para a clínica médica *Vida Saudável*. O mapeamento foi realizado a partir do Diagrama de Entidade-Relacionamento (DER), aplicando as regras de normalização (até a 3ª Forma Normal) e garantindo a integridade referencial por meio de chaves primárias e estrangeiras.

Para fins de testes e homologação, o banco de dados foi hospedado em ambiente de nuvem utilizando o servidor **Somee.com** e gerenciado localmente via **SQL Server Management Studio (SSMS)**.

Insights de criação:

- Criar as tabelas independentes primeiro como CONVENIO, ESPECIALIDADE e FUNCIONARIOS, pois é o mais recomendado para que as demais tabelas tenham as chaves estrangeiras sem precisar usar o comando **ALTER**.

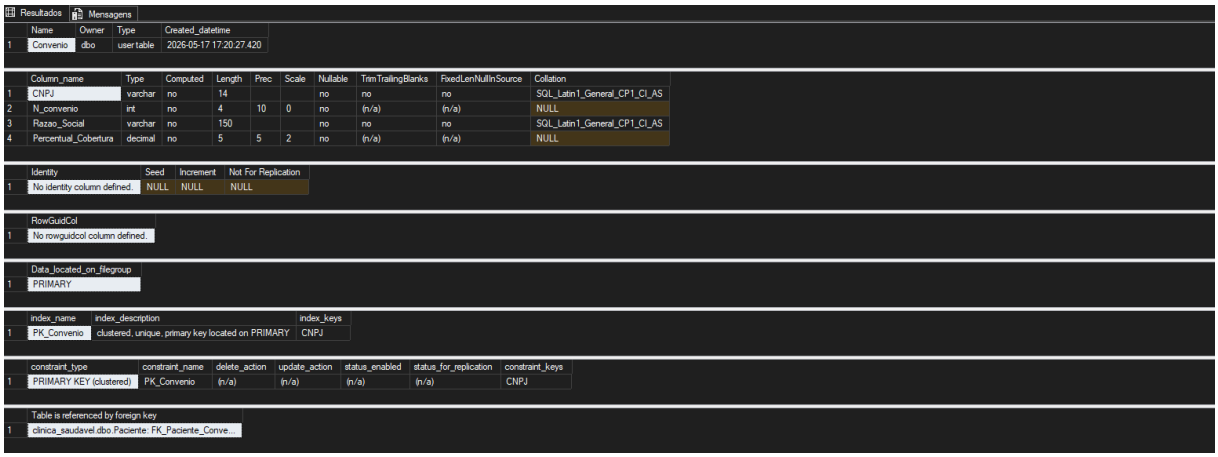
CÓDIGO:

Primeiro criamos o banco de dados que vai suportar as tabelas:

```
CREATE DATABASE
ClinicaSaudavel;
GO
```

Agora criamos as tabelas

```
CREATE TABLE Convenio (
        CNPJ VARCHAR(14) NOT NULL,
        N_convenio INT NOT NULL,
        Razao_Social VARCHAR(150) NOT NULL,
        Percentual_Cobertura DECIMAL(5,2) NOT NULL,
        CONSTRAINT PK_Convenio PRIMARY KEY (CNPJ)
);
```



Resultados | Mensagens

	Name	Owner	Type	Created_datetime
1	Convenio	dbo	user table	2026-05-17 17:20:27.420

	Column_name	Type	Computed	Length	Prec	Scale	Nullable	TrimTrailingBlanks	FixedLenNullInSource	Collation
1	CNPJ	varchar	no	14			no	no	no	SQL_Latin1_General_CP1_CI_AS
2	N_convenio	int	no	4	10	0	no	(n/a)	(n/a)	NULL
3	Razao_Social	varchar	no	150			no	no	no	SQL_Latin1_General_CP1_CI_AS
4	Percentual_Cobertura	decimal	no	5	5	2	no	(n/a)	(n/a)	NULL

	Identity	Seed	Increment	Not For Replication
1	No identity column defined.	NULL	NULL	NULL

	RowGuidCol
1	No rowguidcol column defined.

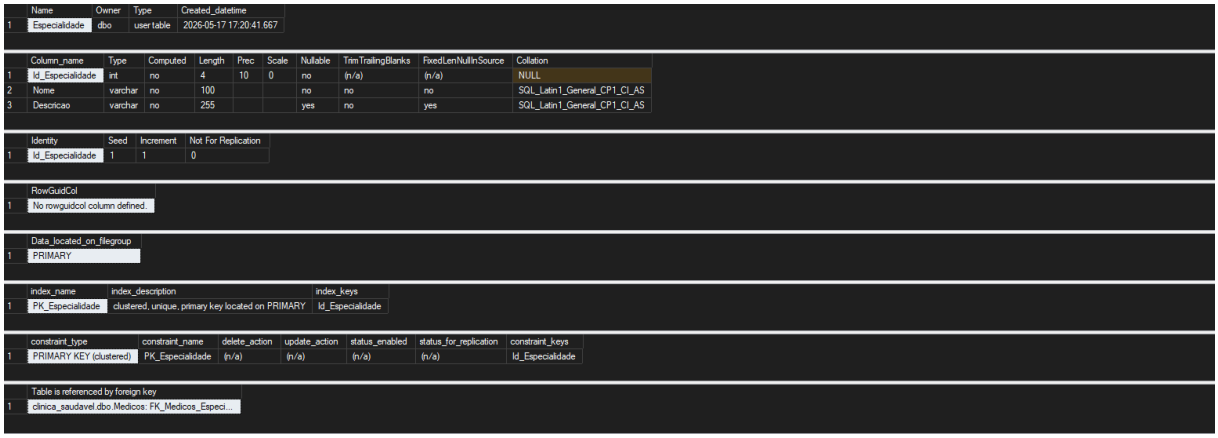
	Data_located_on_filegroup
1	PRIMARY

	index_name	index_description	index_keys
1	PK_Convenio	clustered, unique, primary key located on PRIMARY	CNPJ

	constraint_type	constraint_name	delete_action	update_action	status_enabled	status_for_replication	constraint_keys
1	PRIMARY KEY (clustered)	PK_Convenio	(n/a)	(n/a)	(n/a)	(n/a)	CNPJ

	Table is referenced by foreign key
1	clinica_saudavel.dbo.Paciente: FK_Paciente_Conve...

```
CREATE TABLE Especialidade (
        Id_Especialidade INT IDENTITY(1,1),
        Nome VARCHAR(100) NOT NULL,
        Descricao VARCHAR(255),
        CONSTRAINT PK_Especialidade PRIMARY KEY (Id_Especialidade)
    );
```



	Name	Owner	Type	Created_datetime
1	Especialidade	dbo	user table	2026-05-17 17:20:41.667

	Column_name	Type	Computed	Length	Prec	Scale	Nullable	TrimTrailingBlanks	FixedLenNullInSource	Collation
1	Id_Especialidade	int	no	4	10	0	no	(n/a)	(n/a)	NULL
2	Nome	varchar	no	100			no	no	no	SQL_Latin1_General_CP1_CI_AS
3	Descricao	varchar	no	255			yes	no	yes	SQL_Latin1_General_CP1_CI_AS

	Identity	Seed	Increment	Not For Replication
1	Id_Especialidade	1	1	0

	RowGuidCol
1	No rowguidcol column defined.

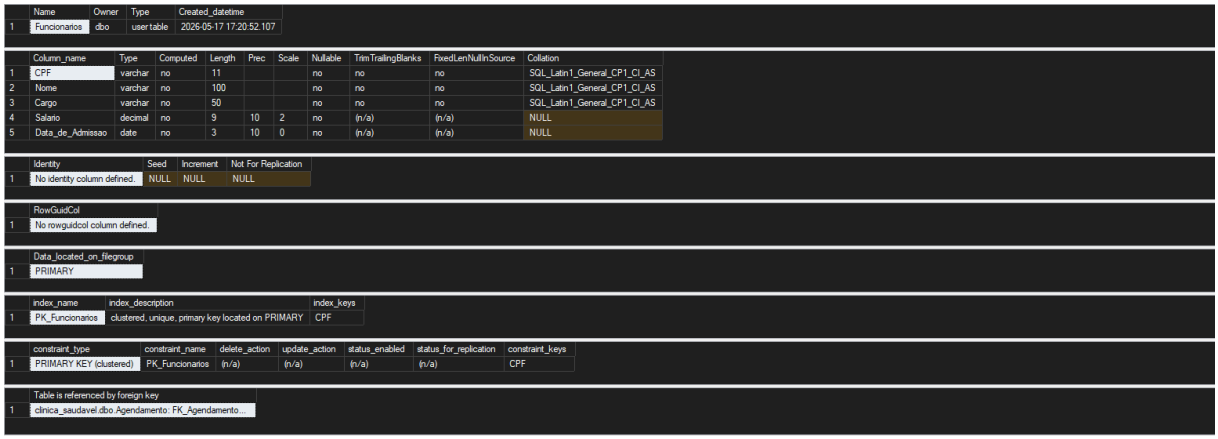
	Data_located_on_filegroup
1	PRIMARY

	index_name	index_description	index_keys
1	PK_Especialidade	clustered, unique, primary key located on PRIMARY	Id_Especialidade

	constraint_type	constraint_name	delete_action	update_action	status_enabled	status_for_replication	constraint_keys
1	PRIMARY KEY (clustered)	PK_Especialidade	(n/a)	(n/a)	(n/a)	(n/a)	Id_Especialidade

	Table is referenced by foreign key
1	clinica_saudavel.dbo.Medicos: FK_Medicos_Especi...

```
CREATE TABLE Funcionarios (
    CPF VARCHAR(11) NOT NULL,
    Nome VARCHAR(100) NOT NULL,
    Cargo VARCHAR(50) NOT NULL,
    Salario DECIMAL(10,2) NOT NULL,
    Data_de_Admissao DATE NOT NULL,
    CONSTRAINT PK_Funcionarios PRIMARY KEY (CPF)
);
```



	Name	Owner	Type	Created_datetime
1	Funcionarios	dbo	user table	2026-05-17 17:20:52.107

	Column_name	Type	Computed	Length	Prec	Scale	Nullable	TrimTrailingBlanks	FixedLenNullInSource	Collation
1	CPF	varchar	no	11			no	no	no	SQL_Latin1_General_CP1_CI_AS
2	Nome	varchar	no	100			no	no	no	SQL_Latin1_General_CP1_CI_AS
3	Cargo	varchar	no	50			no	no	no	SQL_Latin1_General_CP1_CI_AS
4	Salario	decimal	no	9	10	2	no	(n/a)	(n/a)	NULL
5	Data_de_Admissao	date	no	3	10	0	no	(n/a)	(n/a)	NULL

	Identity	Seed	Increment	Not For Replication
1	No identity column defined.	NULL	NULL	NULL

	RowGuidCol
1	No rowguidcol column defined.

	Data_located_on_filegroup
1	PRIMARY

	index_name	index_description	index_keys
1	PK_Funcionarios	clustered, unique, primary key located on PRIMARY	CPF

	constraint_type	constraint_name	delete_action	update_action	status_enabled	status_for_replication	constraint_keys
1	PRIMARY KEY (clustered)	PK_Funcionarios	(n/a)	(n/a)	(n/a)	(n/a)	CPF

	Table is referenced by foreign key
1	clinica_saudavel.dbo.Agendamento: FK_Agendamento...

Após as tabelas independentes, fizemos as tabelas dependentes como **PACIENTE, MEDICO, AGENDAMENTO, PRONTUARIO** e **PRESCRICAO_MEDICA**.

```
CREATE TABLE Paciente (
        CPF VARCHAR(11) NOT NULL,
        Nome VARCHAR(100) NOT NULL,
        RG VARCHAR(20),
        Data_Nasc DATE NOT NULL,
        Sexo CHAR(1) CHECK (Sexo IN ('M', 'F')),
        Email VARCHAR(150),
        CNPJ_Convenio VARCHAR(14),
        Rua VARCHAR(150),
        CEP VARCHAR(8),
        Bairro VARCHAR(50),
        Cidade VARCHAR(50),
        Estado CHAR(2),
        CONSTRAINT PK_Paciente PRIMARY KEY (CPF),
        CONSTRAINT FK_Paciente_Convenio FOREIGN KEY (CNPJ_Convenio) REFERENCES Convenio(CNPJ)
    );
```

1	Paciente	dbo	user table	2026-05-17 17:37:39.953

	Column_name	Type	Computed	Length	Prec	Scale	Nullable	TrimTrailingBlanks	FixedLenNullInSource	Collation
5	Sexo	char	no	1			yes	no	yes	SQL_Latin1_General_CP1_CI_AS
6	Email	varchar	no	150			yes	no	yes	SQL_Latin1_General_CP1_CI_AS
7	CNPJ_Convenio	varchar	no	14			yes	no	yes	SQL_Latin1_General_CP1_CI_AS
8	Rua	varchar	no	150			yes	no	yes	SQL_Latin1_General_CP1_CI_AS
9	CEP	varchar	no	8			yes	no	yes	SQL_Latin1_General_CP1_CI_AS
10	Bairro	varchar	no	50			yes	no	yes	SQL_Latin1_General_CP1_CI_AS
11	Cidade	varchar	no	50			yes	no	yes	SQL_Latin1_General_CP1_CI_AS
12	Estado	char	no	2			yes	no	yes	SQL_Latin1_General_CP1_CI_AS

	Identity	Seed	Increment	Not For Replication
1	No identity column defined.	NULL	NULL	NULL

	RowGuidCol
1	No rowguidcol column defined.

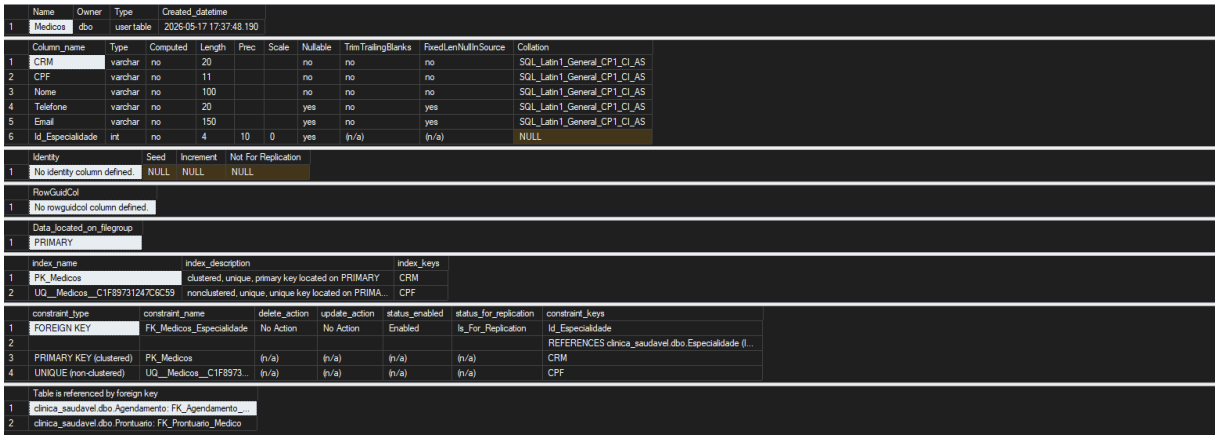
	Data_located_on_filegroup
1	PRIMARY

	index_name	index_description	index_keys
1	PK_Paciente	clustered, unique, primary key located on PRIMARY	CPF

	constraint_type	constraint_name	delete_action	update_action	status_enabled	status_for_replication	constraint_keys
1	CHECK on column Sexo	CK__Paciente__Sexo__3D5E1FD2	(n/a)	(n/a)	Enabled	Is_For_Replication	([Sexo]='F' OR [Sexo]='M')
2	FOREIGN KEY	FK_Paciente_Convenio	No Action	No Action	Enabled	Is_For_Replication	CNPJ_Convenio
3							REFERENCES clinica_...
4	PRIMARY KEY (cluste...	PK_Paciente	(n/a)	(n/a)	(n/a)	(n/a)	CPF

	Table is referenced by foreign key
1	clinica_saudavel.dbo.Agendamento: FK_Agendamento_...
2	clinica_saudavel.dbo.Prontuario: FK_Prontuario_Paciente

```
CREATE TABLE Medicos (
        CRM VARCHAR(20) NOT NULL,
        CPF VARCHAR(11) NOT NULL UNIQUE,
        Nome VARCHAR(100) NOT NULL,
        Telefone VARCHAR(20),
        Email VARCHAR(150),
        Id_Especialidade INT,
        CONSTRAINT PK_Medicos PRIMARY KEY (CRM),
        CONSTRAINT FK_Medicos_Especialidade FOREIGN KEY (Id_Especialidade) REFERENCES Especialidade(Id_Especialidade)
    );
```



	Name	Owner	Type	Created_datetime
1	Medicos	dbo	user table	2026-05-17 17:37:48.190

	Column_name	Type	Computed	Length	Prec	Scale	Nullable	TrimTrailingBlanks	FixedLenNullInSource	Collation
1	CRM	varchar	no	20			no	no	no	SQL_Latin1_General_CP1_CI_AS
2	CPF	varchar	no	11			no	no	no	SQL_Latin1_General_CP1_CI_AS
3	Nome	varchar	no	100			no	no	no	SQL_Latin1_General_CP1_CI_AS
4	Telefone	varchar	no	20			yes	no	yes	SQL_Latin1_General_CP1_CI_AS
5	Email	varchar	no	150			yes	no	yes	SQL_Latin1_General_CP1_CI_AS
6	Id_Especialidade	int	no	4	10	0	yes	(n/a)	(n/a)	NULL

	Identity	Seed	Increment	Not For Replication
1	No identity column defined.	NULL	NULL	NULL

	RowGuidCol
1	No rowguidcol column defined.

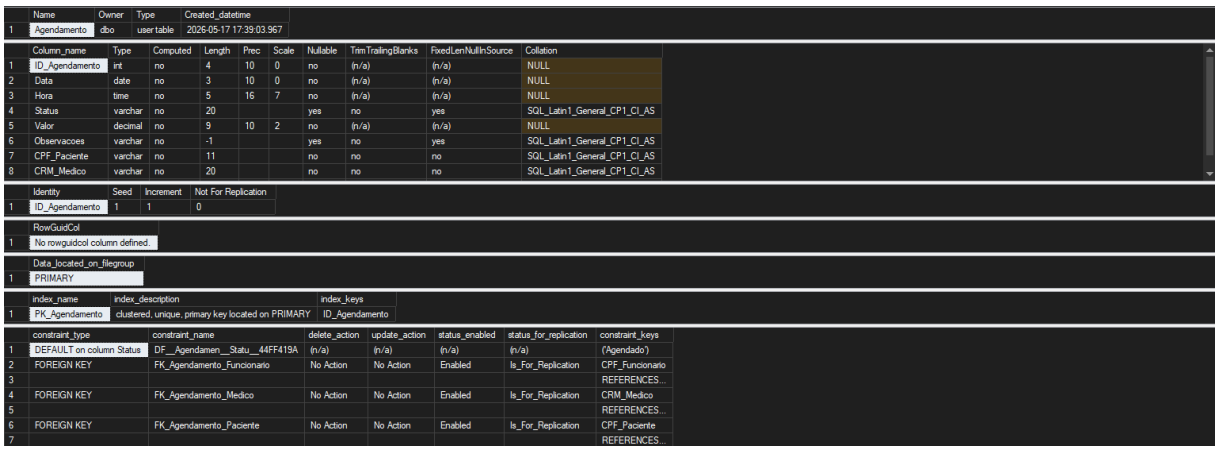
	Data_located_on_filegroup
1	PRIMARY

	index_name	index_description	index_keys
1	PK_Medicos	clustered, unique, primary key located on PRIMARY	CRM
2	UQ__Medicos__C1F89731247C6C59	nonclustered, unique, unique key located on PRIMA...	CPF

	constraint_type	constraint_name	delete_action	update_action	status_enabled	status_for_replication	constraint_keys
1	FOREIGN KEY	FK_Medicos_Especialidade	No Action	No Action	Enabled	Is_For_Replication	Id_Especialidade
2							REFERENCES clinica_saudavel.dbo.Especialidade (I...
3	PRIMARY KEY (clustered)	PK_Medicos	(n/a)	(n/a)	(n/a)	(n/a)	CRM
4	UNIQUE (non-clustered)	UQ__Medicos__C1F8973...	(n/a)	(n/a)	(n/a)	(n/a)	CPF

	Table is referenced by foreign key
1	clinica_saudavel.dbo.Agendamento: FK_Agendamento_...
2	clinica_saudavel.dbo.Prontuario: FK_Prontuario_Medico

```
CREATE TABLE Agendamento (
    ID_Agendamento INT IDENTITY(1,1),
    Data DATE NOT NULL,
    Hora TIME NOT NULL,
    Status VARCHAR(20) DEFAULT 'Agendado',
    Valor DECIMAL(10,2) NOT NULL,
    Observacoes VARCHAR(MAX),
    CPF_Paciente VARCHAR(11) NOT NULL,
    CRM_Medico VARCHAR(20) NOT NULL,
    CPF_Funcionario VARCHAR(11) NOT NULL,
    CONSTRAINT PK_Agendamento PRIMARY KEY (ID_Agendamento),
    CONSTRAINT FK_Agendamento_Paciente FOREIGN KEY (CPF_Paciente) REFERENCES Paciente(CPF),
    CONSTRAINT FK_Agendamento_Medico FOREIGN KEY (CRM_Medico) REFERENCES Medicos(CRM),
    CONSTRAINT FK_Agendamento_Funcionario FOREIGN KEY (CPF_Funcionario) REFERENCES Funcionarios(CPF)
);
```



	Name	Owner	Type	Created_datetime
1	Agendamento	dbo	user table	2026-05-17 17:39:03.967

	Column_name	Type	Computed	Length	Prec	Scale	Nullable	TrimTrailingBlanks	FixedLenNullInSource	Collation
1	ID_Agendamento	int	no	4	10	0	no	(n/a)	(n/a)	NULL
2	Data	date	no	3	10	0	no	(n/a)	(n/a)	NULL
3	Hora	time	no	5	16	7	no	(n/a)	(n/a)	NULL
4	Status	varchar	no	20			yes	no	yes	SQL_Latin1_General_CP1_CI_AS
5	Valor	decimal	no	9	10	2	no	(n/a)	(n/a)	NULL
6	Observacoes	varchar	no	-1			yes	no	yes	SQL_Latin1_General_CP1_CI_AS
7	CPF_Paciente	varchar	no	11			no	no	no	SQL_Latin1_General_CP1_CI_AS
8	CRM_Medico	varchar	no	20			no	no	no	SQL_Latin1_General_CP1_CI_AS

	Identity	Seed	Increment	Not For Replication
1	ID_Agendamento	1	1	0

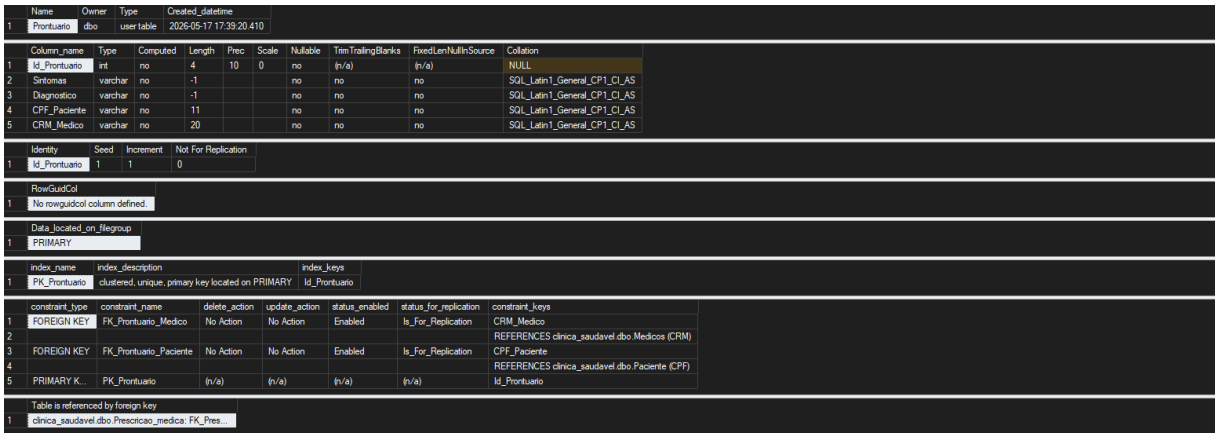
	RowGuidCol
1	No rowguidcol column defined.

	Data_located_on_filegroup
1	PRIMARY

	index_name	index_description	index_keys
1	PK_Agendamento	clustered, unique, primary key located on PRIMARY	ID_Agendamento

	constraint_type	constraint_name	delete_action	update_action	status_enabled	status_for_replication	constraint_keys
1	DEFAULT on column Status	DF__Agendamen__Statu__44FF419A	(n/a)	(n/a)	(n/a)	(n/a)	('Agendado')
2	FOREIGN KEY	FK_Agendamento_Funcionario	No Action	No Action	Enabled	Is_For_Replication	CPF_Funcionario
3							REFERENCES...
4	FOREIGN KEY	FK_Agendamento_Medico	No Action	No Action	Enabled	Is_For_Replication	CRM_Medico
5							REFERENCES...
6	FOREIGN KEY	FK_Agendamento_Paciente	No Action	No Action	Enabled	Is_For_Replication	CPF_Paciente
7							REFERENCES...

```
CREATE TABLE Prontuario (
        Id_Prontuario INT IDENTITY(1,1),
        Sintomas VARCHAR(MAX) NOT NULL,
        Diagnostico VARCHAR(MAX) NOT NULL,
        CPF_Paciente VARCHAR(11) NOT NULL,
        CRM_Medico VARCHAR(20) NOT NULL,
        CONSTRAINT PK_Prontuario PRIMARY KEY (Id_Prontuario),
        CONSTRAINT FK_Prontuario_Paciente FOREIGN KEY (CPF_Paciente) REFERENCES Paciente(CPF),
        CONSTRAINT FK_Prontuario_Medico FOREIGN KEY (CRM_Medico) REFERENCES Medicos(CRM)
    );
```



	Name	Owner	Type	Created_datetime
1	Prontuario	dbo	user table	2026-05-17 17:39:20.410

	Column_name	Type	Computed	Length	Prec	Scale	Nullable	TrimTrailingBlanks	FixedLenNullInSource	Collation
1	Id_Prontuario	int	no	4	10	0	no	(n/a)	(n/a)	NULL
2	Sintomas	varchar	no	-1			no	no	no	SQL_Latin1_General_CP1_CI_AS
3	Diagnostico	varchar	no	-1			no	no	no	SQL_Latin1_General_CP1_CI_AS
4	CPF_Paciente	varchar	no	11			no	no	no	SQL_Latin1_General_CP1_CI_AS
5	CRM_Medico	varchar	no	20			no	no	no	SQL_Latin1_General_CP1_CI_AS

	Identity	Seed	Increment	Not For Replication
1	Id_Prontuario	1	1	0

	RowGuidCol
1	No rowguidcol column defined.

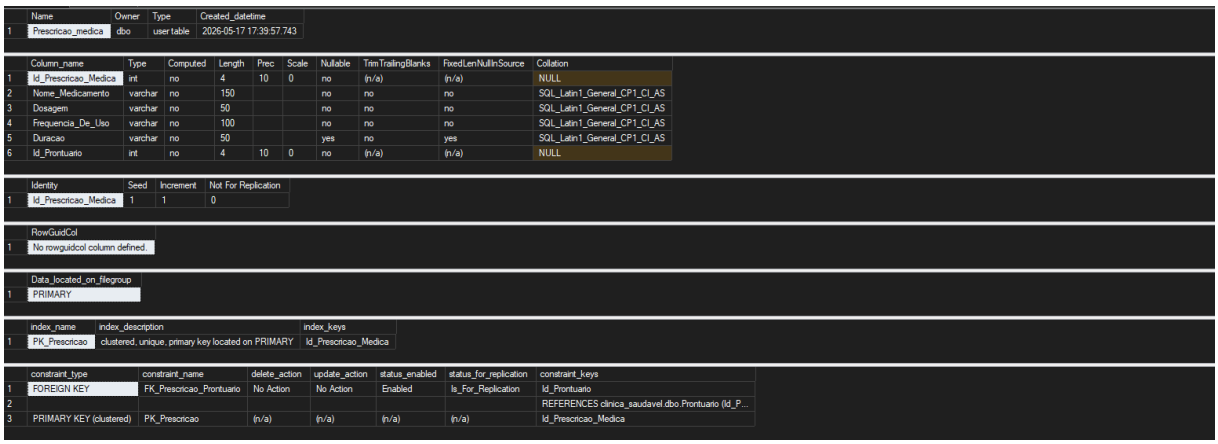
	Data_located_on_filegroup
1	PRIMARY

	index_name	index_description	index_keys
1	PK_Prontuario	clustered, unique, primary key located on PRIMARY	Id_Prontuario

	constraint_type	constraint_name	delete_action	update_action	status_enabled	status_for_replication	constraint_keys
1	FOREIGN KEY	FK_Prontuario_Medico	No Action	No Action	Enabled	Is_For_Replication	CRM_Medico
2							REFERENCES clinica_saudavel.dbo.Medicos (CRM)
3	FOREIGN KEY	FK_Prontuario_Paciente	No Action	No Action	Enabled	Is_For_Replication	CPF_Paciente
4							REFERENCES clinica_saudavel.dbo.Paciente (CPF)
5	PRIMARY K...	PK_Prontuario	(n/a)	(n/a)	(n/a)	(n/a)	Id_Prontuario

	Table is referenced by foreign key
1	clinica_saudavel.dbo.Prescricao_medica: FK_Pres...

```
CREATE TABLE Prescricao_medica (
    Id_Prescricao_Medica INT IDENTITY(1,1),
    Nome_Medicamento VARCHAR(150) NOT NULL,
    Dosagem VARCHAR(50) NOT NULL,
    Frequencia_De_Uso VARCHAR(100) NOT NULL,
    Duracao VARCHAR(50),
    Id_Prontuario INT NOT NULL,
    CONSTRAINT PK_Prescricao PRIMARY KEY (Id_Prescricao_Medica),
    CONSTRAINT FK_Prescricao_Prontuario FOREIGN KEY (Id_Prontuario) REFERENCES Prontuario(Id_Prontuario)
);
```



	Name	Owner	Type	Created_datetime
1	Prescricao_medica	dbo	user table	2026-05-17 17:39:57.743

	Column_name	Type	Computed	Length	Prec	Scale	Nullable	TrimTrailingBlanks	FixedLenNullInSource	Collation
1	Id_Prescricao_Medica	int	no	4	10	0	no	(n/a)	(n/a)	NULL
2	Nome_Medicamento	varchar	no	150			no	no	no	SQL_Latin1_General_CP1_CI_AS
3	Dosagem	varchar	no	50			no	no	no	SQL_Latin1_General_CP1_CI_AS
4	Frequencia_De_Uso	varchar	no	100			no	no	no	SQL_Latin1_General_CP1_CI_AS
5	Duracao	varchar	no	50			yes	no	yes	SQL_Latin1_General_CP1_CI_AS
6	Id_Prontuario	int	no	4	10	0	no	(n/a)	(n/a)	NULL

	Identity	Seed	Increment	Not For Replication
1	Id_Prescricao_Medica	1	1	0

	RowGuidCol
1	No rowguidcol column defined.

	Data_located_on_filegroup
1	PRIMARY

	index_name	index_description	index_keys
1	PK_Prescricao	clustered, unique, primary key located on PRIMARY	Id_Prescricao_Medica

	constraint_type	constraint_name	delete_action	update_action	status_enabled	status_for_replication	constraint_keys
1	FOREIGN KEY	FK_Prescricao_Prontuario	No Action	No Action	Enabled	Is_For_Replication	Id_Prontuario
2							REFERENCES clinica_saudavel.dbo.Prontuario (Id_P...
3	PRIMARY KEY (clustered)	PK_Prescricao	(n/a)	(n/a)	(n/a)	(n/a)	Id_Prescricao_Medica

Todas as tabelas criadas e listadas:

